

Donghyung Lee

DevOps Engineer

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Portfolio: somaz.blog/portfolio

Introduction

Hello, I am Donghyung Lee, a DevOps Engineer who designs infrastructure from the ground up and leads its transitions.

Rather than maintaining what was already there, I have mostly led greenfield builds and large-scale migrations at business turning points. Starting with an OpenStack-based private cloud for financial and public-sector clients, I went on to lead the infrastructure side of a company-wide AWS → GCP migration at a startup, cutting cloud cost 50%. Today I am the sole owner of the AWS + on-premises hybrid infrastructure at a game studio, where I cut monthly infrastructure cost 20% and solely built and launched the EKS production environment for a new game.

My main strength is CI/CD automation, with observability layered on top for production stability. I have built CI/CD pipelines on GitLab CI, GitHub Actions and Jenkins with ArgoCD, and today on Kubernetes I fully automated build and deployment with GitLab CI and ArgoCD GitOps, cutting deployment time 67% (30min → 10min). New components are 100% codified with Terraform and Ansible, and on top of that a 3-signal observability stack tying logs, metrics and distributed tracing together cut the time it takes to notice an incident and find its cause (MTTR).

Beyond that, I build in-house DevOps tools myself to remove operational bottlenecks, and I have worked on AI/ML workload infrastructure as well. At the startup I standardized and scaled an image-generation GPU fleet across 10 in-house boxes, GCP A100, and 20 external-IDC servers; today I run a docker compose-based in-house LLM inference service on an on-prem GPU server, keeping a coding model resident and open to the dev team.

As the sole owner of the infrastructure, I set myself the principle that systems and documents — not one person — should do the work. I believe infrastructure is finished not when it ships but when a fellow developer has a guide they can use on their own, and I keep building environments where solid automation lets the whole team move fast with confidence.

Strengths

Asking 'Why' First

- Habit of asking 'why' first — define the business problem before picking the tool

Systematic Documentation & Knowledge Sharing

- Document and share every task, building organizational knowledge assets

Record-Based Continuous Improvement

- Postmortem-record every incident, infra change, and troubleshooting — preventing recurrence, driving improvement

Full-Stack Infrastructure Experience

- End-to-end infra across cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP, OpenStack
Container Orchestration	Kubernetes, Docker, Cilium CNI, NGINX Gateway Fabric / Gateway API, Karpenter
Infrastructure as Code	Terraform, Ansible, Helm, Helmfile
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, Argo Rollouts, Jenkins, GitOps
Monitoring & Observability	PLG Stack (Prometheus/Loki/Grafana), ELK Stack, EFK Stack, ECK Operator, Fluent-bit, Thanos, OpenTelemetry, Grafana Tempo
Security & IAM	Keycloak (Operator, OIDC IdP), Vaultwarden, GitLab SSO (OIDC), External Secrets Operator, Sealed Secrets
Automation	Go, Python, Shell Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Professional Experience

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

Sole DevOps infrastructure engineer at a game studio, designing, building, and operating AWS-based cloud and on-premises servers. Codified infrastructure with Terraform/Ansible (100% IaC for new components) and documented operations via automation scripts and postmortems.

- Cut monthly infra cost 20% — sole design and build of the AWS + on-prem hybrid, 100% IaC on new components (Terraform/Ansible)
- Sole-built the launch infrastructure for a new game, expanding a single on-prem dev cluster into an on-prem + AWS multi-cluster. EKS platform 100% GitOps, Karpenter spot autoscaling, a move to EKS Auto Mode cutting compute Terraform management points 83% (102 → 17), and a WireGuard site-to-site VPN giving on-prem a private route
- Built Metrics · Traces · Logs 3-signal observability — log search on one Kibana interface (a 3-stage redesign: ELK → EFK → ECK Operator), OpenTelemetry distributed tracing (OTel Collector · Tempo) and kube-prometheus-stack
- Extended the EFK log pipeline into a product-analytics platform — ES Transform aggregating events per user and Kibana serving D+1~D+30 retention and cohorts, with no commercial analytics SaaS
- Automated manual build-and-deploy with GitLab CI + ArgoCD pull-GitOps, cutting deployment time 67% (30min → 10min); shared toolchain images took CI provisioning from 5min to 10s, plus Argo Rollouts Canary and infra automation ported to Python
- Consolidated multiple ingress-nginx instances onto NGINX Gateway Fabric, central SSO for Harbor · ArgoCD · Vaultwarden via Keycloak Operator, plaintext DB passwords removed via External Secrets Operator, static SSH keys removed via IRSA/Pod Identity · SSM
- Built solo over 6 weeks a self-service portal placing 7 internal apps behind a single OIDC authentication gate, plus 5 operations services (accounts · Keycloak · Harbor · Elasticsearch · VPN) wired to 10 internal account systems — operational requests that had to pass through one engineer are now handled by each team directly
- Diagnosed the bottleneck in which GitLab Runner build I/O spikes drove second-scale latency on HDD-backed etcd and the game DB, moved the control plane and DB worker to SSD and isolated the build workload onto dedicated physical hardware — game DB COMMIT latency normalized from 9~14s to the millisecond range, with zero in-game connection incidents during builds

Key Achievements

- 20% Monthly Infra Cost Reduction (savings reinvested in new projects)
- Greenfield EKS Sole-Built Production Launch Infra (single on-prem dev cluster → on-prem + AWS multi-cluster, serving real traffic since the 2026.07 soft launch)
- 67% Deployment Time Reduction (30min → 10min, faster market response)

NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Led the large-scale AWS → GCP cloud migration at a game and blockchain startup, and built GitOps-based CI/CD pipelines.

- AWS → GCP large-scale migration (Shared VPC + Workload Identity Federation keyless GitHub Actions auth, subdomain-delegated DNS cutover during a maintenance window, Cloud Armor + Cloud CDN, self-hosted NFS replacing Filestore)
- Built GitOps CI/CD pipeline on GitLab CI · GitHub Actions · Jenkins · ArgoCD
- Built game/blockchain data pipeline (MongoDB · CloudSQL · GA · Dune → BigQuery → Google Sheets, Cloud Functions + Scheduler + Dataflow, Terraform IaC) — removed 2-3 hours of daily manual work, 95% automated
- Unified monitoring (PLG Stack: Prometheus · Loki · Grafana) and certificate automation (Cert-Manager + Let's Encrypt)
- Designed & built GPU infrastructure for image-generation AI workloads — 10 in-house boxes, GCP A100, 20 external-IDC servers

Key Achievements

- 50% Cloud Cost Reduction (Fargate → GKE node pools + MSP discount)
- 95% Data Collection Automation (0% data gap rate)

Iorchard

2022.04 ~ 2023.03

Infra Engineer

Designed and built PaaS (Kubernetes + OpenStack + Ceph) private clouds for finance/public-sector clients; led operations training.

- Customized PaaS solution design and construction — delivered to 5 clients (Kubernetes HA + OpenStack + Ceph)
- DP (Delivery Partner) Kubernetes operations training program design and delivery (6 months)
- Kubernetes certificate validity extension automation script (1yr → 10yr)
- Operated and optimized OpenStack and Ceph storage clusters, with regular cluster inspection and troubleshooting
- Ansible-based server provisioning automation achieving zero configuration drift

Key Achievements

- 5 Cases Client PaaS Solution Deployments (Stable HA-based operation)
 - 10x Certificate Renewal Cycle Extension (Annual → Once per 10 years)
 - 50hrs Annual Ops Savings (Across all 10 client sites then in operation)
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ETech System

2022.01 ~ 2022.04

System Engineer

- Hardware Server and SAN storage construction
- Transitioned to DevOps field for career growth

Opensource Projects

Kubernetes CRD

[k8s-namespace-sync](#) · [helios-lb](#) · [network-policy-generator](#) · [kube-drift](#)

Helm Charts

[nginx-gateway-cr](#) · [elasticsearch-eck](#) · [kibana-eck](#) · [certmanager-letsencrypt](#) · [karpenter-cr](#) · [metallb-cr](#) · [prometheus-cr](#) · [external-secrets-cr](#) · [keycloak-cr](#) · [keycloak-operator](#) · [aws-storageclass](#) · [mysql](#) · [postgresql](#) · [redis](#) · [buildkit](#) · [ghost](#) · [unity-mcp-server](#)

GitHub Actions

[Compress Decompress Action](#) · [Go Git Commit Action](#) · [Multi Git Mirror Action](#) · [Image Tag Updater](#) · [Kube Diff Action](#) · [Helm Chart Release Action](#) · [Go Kubebuilder Test Action](#) · [Helm OCI Push](#) · [Helm Kustomize Lint Action](#) · [Major Tag Action](#) · [Go Changelog Generator](#) · [Kind E2E Test Action](#) · [Go Docker Action CI](#) · [Contributors Action](#) · [Env Output Setter](#) · [Ansible Molecule Test Action](#) · [Ansible Galaxy Publish Action](#) · [Ternary Operator](#) · [Commit Info Extractor](#)

Ansible Galaxy

[Ansible K8s IAC Tool](#) · [Ansible Kubectl Krew](#) · [Ansible User Management](#)

DevOps Tools

[bash-pilot](#) · [git-bridge](#) · [static-file-server](#) · [kube-diff](#) · [kube-events](#) · [kube-ctx](#) · [helm-upgrade-sync](#) · [helm-chart-kit](#) · [kubeconfig-generator](#) · [slack-qr-bot](#) · [script-collection](#)

Chrome Extensions

[Dev Toolkit](#) · [QRify](#)

Package Distribution

[homebrew-tap](#) · [scoop-bucket](#) · [krew-index](#)

Claude Code Plugins

[census](#) · [atlas](#) · [sensitive-guard](#) · [release-guards](#) · [session-continuity](#) · [shell-portability](#) · [doc-mirror](#)

Education

Chungnam National University – Public Administration (Transfer)

2018.03 ~ 2020.08

National Institute for Lifelong Education – Business Administration (Academic Credit Bank)

2016.09 ~ 2018.02

Certifications

Engineer Information Processing

2021.11

AWS Certified Solution Architect – Associate (Expired)

2021.11

Network Administrator Level 2

2021.10

Linux Master Level 2

2021.10

Computer Specialist Level 1

2021.04

TOEIC Speaking Test – 130/Intermediate Mid 3 (Expired)

2021.02

Study

[Tech blog \(somaz.blog, English\)](#) — [DevOps](#) · [Kubernetes](#) · [IaC writing for global readers](#)

2025.01 ~ 현재

[Tech blog \(somaz.tistory.com, Korean\)](#) — [ongoing DevOps](#) · [Kubernetes](#) · [IaC learning and operations notes](#)

2023.04 ~ 현재

[T101\(Terraform\) Study](#)

2023.08 ~ 2023.10

[AEWS\(EKS\) Study](#)

2023.04 ~ 2023.06

[PKOS\(Kubernetes\) Study](#)

2023.01 ~ 2023.02

[Cloud Security Training](#)

2022.02 ~ 2022.04

[Security Solution Operation Specialist Training Program](#)

2021.07 ~ 2021.12